

1. What does the precision of a classifier refer to?

1 Marks

The percentage of predicted positive instances that are actually positive.

The percentage of actual positive instances correctly identified.

The total number of true positives.

The total number of false negatives.

2. What does the recall of a classifier measure?

1 Marks

The percentage of actual positive instances correctly identified.

The percentage of predicted positive instances that are actually positive.

The total number of true positives.

The total number of false negatives.

3. Which ensemble method combines multiple classifiers by training each classifier on a different bootstrap sample of the data?

1 Marks

Stacking

Boosting

Bagging

Voting

4. Which of the following represents a classifier?

1 Marks

Decision tree

Regression model

Clustering algorithm

Association rule

5. What is a decision tree used for in Machine Learning?

1 Marks

Data cleaning

Predicting categorical class labels

Predicting numerical values

Data visualization

6. What is the outcome of bagging in ensemble methods?

1 Marks

Overfitting

Decreased accuracy

Improved classification performance

Reduction in training data

7. Which of the following is a decision tree algorithm?

1 Marks

CART

Naive Bayes

k-means

Apriori

8. In Bayesian classification, what is $P(H|X)$?

1 Marks

Evidence

Prior probability

Likelihood

Posterior probability

9. What is the F1-score used for?

1 Marks

Measuring accuracy

Measuring precision only

Measuring recall only

Balancing precision and recall

10. What is the purpose of the holdout method?

1 Marks

To train a model on the entire dataset

To divide the data into training and test sets

To reduce the size of the dataset

To select the best attributes

11. What is the main advantage of using boosting?

1 Marks

Reduces the number of models

Simplifies the model

Increases accuracy

Uses fewer training data

12. What is the formula of error rate?

1 Marks

$1/\text{accuracy}(M)$

$1+\text{accuracy}(M)$

$1-\text{accuracy}(M)$

None of the options given

13. What does the confusion matrix help evaluate in a classification model?

1 Marks

Model accuracy

Class frequencies

Test set partitioning

Error rate

14. Give the formula for Precision.

1 Marks

None of the options given

$TP / (TP - FP)$

$TF / (TP + FP)$

$TP / (TP + FP)$

15. Which one of the following is the method to construct Random Forest?

1 Marks

Boosting

Bootstrap Aggregation (Bagging) combined with Random Feature Selection.

Stacking

Bagging with Decision Trees

16. What is the error rate of a classifier?

1 Marks

The fraction of incorrect predictions

The fraction of correct predictions

The fraction of positive predictions

The fraction of negative predictions

17. What is the purpose of using a confusion matrix?

1 Marks

To visualize model accuracy

To compare different classifiers

To evaluate the performance of a classification model

To reduce data dimensionality

18. Which of the following statements about SVC is TRUE?

1 Marks

SVC can use the kernel trick to classify non-linearly separable data

The number of support vectors increases with increasing margin

SVC works only for linearly separable data

SVC cannot handle imbalanced data

19. What is the primary factor that determines the classification in K-Nearest Neighbors (KNN)?

1 Marks

Majority vote among the k-nearest neighbors

Decision boundary

Number of support vectors

Hyperplane margin

20. What type of learning does KNN belong to?

1 Marks

Instance-based learning

Supervised learning

Reinforcement learning

Unsupervised learning

21. What type of classifiers does AdaBoost primarily use?

1 Marks

Support Vector Machines

Deep neural networks

K-Nearest Neighbors

Weak learners, typically decision stumps

22. What is the main difference between Bagging and Boosting?

1 Marks

Bagging reduces variance, while Boosting reduces bias

Bagging and Boosting both reduce variance

Boosting randomly selects training samples, while Bagging assigns weights to samples

Boosting works only with decision trees, while Bagging does not

23. Which of the following is an example of a Bagging algorithm?

1 Marks

Random Forest

AdaBoost

Gradient Boosting

XGBoost

24. How does Boosting train weak learners?

1 Marks

Sequentially, with each new learner correcting the mistakes of the previous one

Independently and in parallel

By using the same data for each learner

By averaging predictions from multiple models

25. Which technique is typically better when dealing with large datasets and high variance?

1 Marks

Boosting

Bagging

Both perform equally well

Neither is useful for large datasets

26. Which Scikit-Learn module contains the Support Vector Classifier (SVC)?

1 Marks

`sklearn.neighbors`

`sklearn.svm`

`sklearn.tree`

`sklearn.ensemble`

27. Which Scikit-Learn class is used to implement Bagging?

1 Marks

`sklearn.ensemble.BoostingClassifier`

`sklearn.ensemble.BaggingClassifier`

`sklearn.tree.DecisionTreeClassifier`

`sklearn.ensemble.RandomForestClassifier`

28. Which Scikit-Learn function is used to compute the ROC curve?

1 Marks

`sklearn.metrics.roc_curve`

`sklearn.metrics.confusion_matrix`

`sklearn.metrics.precision_recall_curve`

`sklearn.metrics.f1_score`

29. After successfully training a Random Forest model, you want to identify which features had the biggest impact on the predictions:

Which of the following lines of code correctly accesses the calculated feature importances in Scikit-Learn?

1 Marks

`importances = model.get_weights()`

`importances = model.coef_`

`importances = model.feature_importances_`

`importances = model.importance_scores()`

30. You load a raw CSV file into a pandas DataFrame and immediately attempt to fit a

```
model: import pandas as pd
from sklearn.linear_model import LinearRegression # X_train contains 5 missing values (NaNs) scattered across the columns
model = LinearRegression()
model.fit(X_train, y_train) What will happen when you execute the fit method?
```

1 Marks

Scikit-Learn will automatically impute the missing values with the column mean.

Scikit-Learn will silently drop the 5 rows containing missing values.

The code will crash and throw a `ValueError` because standard Scikit-Learn models cannot natively handle missing data.

The model will train, but the coefficients for the columns with NaNs will be set to zero.

1. What is the primary goal of classification in Machine Learning?

1 Marks

To summarize and describe the main characteristics of a dataset.

To predict a categorical class label for new, unseen data.

To group similar data instances together.

To identify outliers and anomalies in a dataset.

2. Which decision tree algorithm uses the Gini index as an attribute selection measure?

1 Marks

CART

ID3

C4.5

k-Nearest Neighbors

3. What theorem forms the basis of Bayesian classification?

1 Marks

Bayes' Theorem

Central Limit Theorem

Law of Large Numbers

Pythagorean Theorem

4. Which of the following is an example of a decision tree induction algorithm?

1 Marks

Apriori

SVM

K-means

ID3

5. Which of the following is used by ID3 for attribute selection?

1 Marks

Information gain

Gini index

Gain ratio

Entropy

6. What is the outcome of bagging in ensemble methods?

1 Marks

Overfitting

Decreased accuracy

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Reduction in training data

7. What is the bootstrap method used for?

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Reducing model complexity

Data cleaning

Sampling with replacement

Splitting data without replacement

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1 Marks

CART

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9. What does the term 'ensemble' refer to in data mining?

1 Marks

A group of models

A single model

A type of data

A decision tree

10. Which of the following is considered by gini index for each attribute?

1 Marks

Binary split

Ternary split

Singular Split

None of the options given

11. What does entropy measure in the context of decision trees?

1 Marks

Randomness or impurity in data

Data size

Data distribution

Accuracy of the model

12. What does specificity measure in a classifier?

1 Marks

The fraction of true negatives among all negative predictions

The fraction of true positives among all positive predictions

The fraction of false positives among all negative predictions

The fraction of false negatives among all positive predictions

13. What is the formula of error rate?

1 Marks

$1/\text{accuracy}(M)$

$1+\text{accuracy}(M)$

$1-\text{accuracy}(M)$

None of the options given

14. Formula for bayes theorem: $P(H|X) =$

1 Marks

$(P(X|H) * P(H)) / P(X)$

$(P(X|H) + P(H)) / P(X)$

$(P(X|H) * P(X)) / P(H)$

None of the options given

15. Give the formula for Specificity.

1 Marks

TF / P

TP / P

TP / F

TN/N

16. Give the formula for $F\beta$ score.

1 Marks

$((1+\beta) * \text{precision} * \text{recall}) / (\beta * \text{precision} + \text{recall})$

$((1+\beta^2) * \text{precision} * \text{recall}) / (\beta^2 * \text{precision} + \text{recall})$

$((1+\beta) * \text{precision} * \text{recall}) / (\beta^2 + \text{precision} + \text{recall})$

None of the options given

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Correctly labeled positive instances

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20. What is the main objective of Support Vector Classification (SVC)?

1 Marks

Minimize the number of support vectors

Minimize intra-cluster distance

Maximize the margin between classes

Find the most complex decision boundary

21. What is the role of the C parameter in SVC?

1 Marks

Defines the shape of the kernel

Determines the number of nearest neighbors

Controls the trade-off between margin width and classification error

Increases model bias

22. What type of learning does KNN belong to?

1 Marks

Instance-based learning

Supervised learning

Reinforcement learning

Unsupervised learning

23. How does AdaBoost assign weights to training samples?

1 Marks

Randomly assigns weights to samples
Assigns equal weights to all samples
Reduces the weight of misclassified samples
Increases the weight of misclassified samples

24. What type of machine learning algorithm is AdaBoost?

1 Marks

An ensemble learning algorithm

A clustering algorithm

A deep learning algorithm

A reinforcement learning algorithm

25. Which of the following is an example of a Bagging algorithm?

1 Marks

Random Forest

AdaBoost

Gradient Boosting

XGBoost

26. Which class in Scikit-Learn is used for implementing a Decision Tree classifier?

1 Marks

`sklearn.tree.DecisionTreeRegressor`

`sklearn.tree.DecisionTreeClassifier`

`sklearn.ensemble.DecisionTree`

`sklearn.neighbors.DecisionTree`

27. Which Scikit-Learn module contains the Support Vector Classifier (SVC)?

1 Marks

`sklearn.neighbors`

`sklearn.svm`

`sklearn.tree`

`sklearn.ensemble`

28. Which Scikit-Learn function is used to visualize a confusion matrix?

1 Marks

`sklearn.metrics.ConfusionMatrixDisplay`

`sklearn.metrics.plot_roc_curve`

`sklearn.metrics.show_confusion_matrix`

`matplotlib.confusion_matrix`

29. Review the following basic instantiation of a Random Forest: from sklearn.ensemble
import RandomForestClassifier rf = RandomForestClassifier(n_estimators=10)

What does the `n_estimators=10` parameter specifically instruct Scikit-Learn to do?

1 Marks

Restrict the maximum depth of the resulting tree to 10 hierarchical levels.

Use only the 10 most statistically important features from the training dataset.

Train an ensemble consisting of exactly 10 separate decision trees.

Split the dataset into 10 folds for internal cross-validation.

30. You are evaluating a model and see the following two approaches:

from sklearn.model_selection import cross_val_predict # Approach A

model.fit(X, y)

pred_A = model.predict(X) # Approach B

pred_B = cross_val_predict(model, X, y, cv=5)

What is the defining difference between the predictions in `pred_A` and `pred_B`?

1 Marks

`pred_B` predicts probabilities, while `pred_A` predicts continuous integers.

In `pred_B`, each data point's prediction was generated by a model that was never trained on that specific data point.

`pred_A` is used for Regression, while `pred_B` is strictly for Classification.

There is no difference; they yield the exact same numerical array.

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To summarize and describe the main characteristics of a dataset.

To predict a categorical class label for new, unseen data.

To group similar data instances together.

To identify outliers and anomalies in a dataset.

2. What are the two main steps involved in classification?

1 Marks

None of these.

Data collection and data cleaning

Model building and model application

Data preprocessing and pattern discovery

3. What do the inner nodes, edges, and leaf nodes of a decision tree represent, respectively?

1 Marks

Attribute, test on attribute, class label

Class label, test on attribute, attribute

Test on attribute, class label, attribute

Class label, attribute, test on attribute

4. Which decision tree algorithm uses the Gini index as an attribute selection measure?

1 Marks

CART

ID3

C4.5

k-Nearest Neighbors

5. What theorem forms the basis of Bayesian classification?

1 Marks

Bayes' Theorem

Central Limit Theorem

Law of Large Numbers

Pythagorean Theorem

6. What does the precision of a classifier refer to?

1 Marks

The percentage of predicted positive instances that are actually positive.

The percentage of actual positive instances correctly identified.

The total number of true positives.

The total number of false negatives.

7. What is a key characteristic of the Random Forest ensemble method?

1 Marks

It is specifically designed for handling missing values.

It uses a single, highly optimized decision tree.

It does not require any training data.

It combines multiple decision trees, each trained on a different random subset of features.

8. Which of the following represents a classifier?

1 Marks

Decision tree

Regression model

Clustering algorithm

Association rule

9. What does a leaf node in a decision tree represent?

1 Marks

A class label

An attribute

A test on an attribute

A split in the data

10. What does the area under the ROC curve represent?

1 Marks

Model complexity

Data size

Model accuracy

Data distribution

11. Which of the following is a decision tree algorithm?

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CART

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12. What does the term 'ensemble' refer to in data mining?

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A group of models

A single model

A type of data

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Accuracy of the model

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1 Marks

Evidence

Prior probability

Likelihood

Posterior probability

15. What is the F1-score used for?

1 Marks

Measuring accuracy

Measuring precision only

Measuring recall only

Balancing precision and recall

16. What is a random forest in classification?

1 Marks

A combination of decision trees

A single decision tree

A linear classifier

A single Naive Bayes model

17. What does the confusion matrix help evaluate in a classification model?

1 Marks

Model accuracy

Class frequencies

Test set partitioning

Error rate

18. Give the formula for F_β score.

1 Marks

$((1+\beta) * \text{precision} * \text{recall}) / (\beta * \text{precision} + \text{recall})$

$((1+\beta^2) * \text{precision} * \text{recall}) / (\beta^2 * \text{precision} + \text{recall})$

$((1+\beta) * \text{precision} * \text{recall}) / (\beta^2 + \text{precision} + \text{recall})$

None of the options given

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1 Marks

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Minimize intra-cluster distance

Maximize the margin between classes

Find the most complex decision boundary

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1 Marks

SVC can use the kernel trick to classify non-linearly separable data

The number of support vectors increases with increasing margin

SVC works only for linearly separable data

SVC cannot handle imbalanced data

22. Which factor has the most significant impact on KNN's performance?

1 Marks

The activation function used

The number of support vectors

The number of output classes

The choice of k value and distance metric

23. What type of classifiers does AdaBoost primarily use?

1 Marks

Support Vector Machines

Deep neural networks

K-Nearest Neighbors

Weak learners, typically decision stumps

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Bagging reduces variance, while Boosting reduces bias

Bagging and Boosting both reduce variance

Boosting randomly selects training samples, while Bagging assigns weights to samples

Boosting works only with decision trees, while Bagging does not

26. Which of the following is an example of a Bagging algorithm?

1 Marks

Random Forest

AdaBoost

Gradient Boosting

XGBoost

27. Which ensemble learning method is more prone to overfitting?

1 Marks

Boosting, because it focuses on misclassified examples

Bagging, because it creates multiple models

Both Bagging and Boosting equally

Neither Bagging nor Boosting

28. Which Scikit-Learn class is used to implement Bagging?

1 Marks

`sklearn.ensemble.BoostingClassifier`

`sklearn.ensemble.BaggingClassifier`

`sklearn.tree.DecisionTreeClassifier`

`sklearn.ensemble.RandomForestClassifier`

29. Which Scikit-Learn function is used to generate a confusion matrix?

1 Marks

`sklearn.metrics.accuracy_score`

`sklearn.metrics.confusion_matrix`

`sklearn.metrics.classification_report`

`sklearn.metrics.roc_auc_score`

30. Which Scikit-Learn function is used to compute the ROC curve?

1 Marks

`sklearn.metrics.roc_curve`

`sklearn.metrics.confusion_matrix`

`sklearn.metrics.precision_recall_curve`

`sklearn.metrics.f1_score`

1. In the context of classification, what is a classifier?

1 Marks

A model derived from data used to predict class labels.

An algorithm that identifies patterns in data.

A measure of the accuracy of a prediction.

A technique for visualizing data.

2. What do the inner nodes, edges, and leaf nodes of a decision tree represent, respectively?

1 Marks

Attribute, test on attribute, class label

Class label, test on attribute, attribute

Test on attribute, class label, attribute

Class label, attribute, test on attribute

3. What theorem forms the basis of Bayesian classification?

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SVM

K-means

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Model complexity

Data size

Model accuracy

Data distribution

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7. What does the term 'ensemble' refer to in data mining?

1 Marks

A group of models

A single model

A type of data

A decision tree

8. Which of the following is considered by gini index for each attribute?

1 Marks

Binary split

Ternary split

Singular Split

None of the options given

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1 Marks

Measuring accuracy

Measuring precision only

Measuring recall only

Balancing precision and recall

10. Which of the following is true for ensemble methods?

1 Marks

They combine multiple models to improve performance

They reduce model complexity

They use a single model for predictions

They are always based on decision trees

11. What is the main advantage of using boosting?

1 Marks

Reduces the number of models

Simplifies the model

Increases accuracy

Uses fewer training data

12. What is the formula of error rate?

1 Marks

$(FP+FN)/(P+N)$

$(FP-FN)/(P+N)$

$(FP+FN)/(P+N)$

$(FP-FN)/(P-N)$

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1 Marks

$(P(X|H) * P(H)) / P(X)$

$(P(X|H) + P(H)) / P(X)$

$(P(X|H) * P(X)) / P(H)$

None of the options given

14. Give the formula for Specificity.

1 Marks

TF / P

TP / P

TP / F

TN/N

15. Give the formula for Recall.

1 Marks

None of the options given

$TP / (TP - FP)$

$TP / (TP + FP)$

$TP / (TP + FN)$

16. Give the formula for $F\beta$ score.

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$((1+\beta) * \text{precision} * \text{recall}) / (\beta * \text{precision} + \text{recall})$

$((1+\beta^2) * \text{precision} * \text{recall}) / (\beta^2 * \text{precision} + \text{recall})$

$((1+\beta) * \text{precision} * \text{recall}) / (\beta^2 + \text{precision} + \text{recall})$

None of the options given

17. What is the term for the percentage of correctly classified instances in a classification model?

1 Marks

Accuracy

Precision

Recall

Sensitivity

18. What is true positive (TP) in a confusion matrix?

1 Marks

Correctly labeled positive instances

Incorrectly labeled positive instances

Incorrectly labeled negative instances

Correctly labeled negative instances

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Boosting gives more weight to misclassified instances

Boosting combines models with equal weights

Boosting reduces model size

Boosting uses only one model

20. What is the role of the C parameter in SVC?

1 Marks

Defines the shape of the kernel

Determines the number of nearest neighbors

Controls the trade-off between margin width and classification error

Increases model bias

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1 Marks

Majority vote among the k-nearest neighbors

Decision boundary

Number of support vectors

Hyperplane margin

23. What is the function of the kernel trick in SVC?

1 Marks

Map the input space into a higher-dimensional space to make data linearly separable

Reduce the number of features

Speed up model training

Reduce overfitting

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1 Marks

The activation function used

The number of support vectors

The number of output classes

The choice of k value and distance metric

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Assigns equal weights to all samples

Reduces the weight of misclassified samples

Increases the weight of misclassified samples

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1 Marks

AdaBoost assigns different weights to samples, while Bagging uses bootstrapping

Bagging focuses on reducing bias, while AdaBoost reduces variance

Bagging uses weak learners, while AdaBoost does not

AdaBoost requires deep learning models

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1 Marks

Attribute, test on attribute, class label

Class label, test on attribute, attribute

Test on attribute, class label, attribute

Class label, attribute, test on attribute

3. Which of the following decision tree algorithms is based on the information gain measure?

1 Marks

k-Nearest Neighbors

CART

C4.5

ID3

4. In Naive Bayes, what does $P(H|X)$ represent?

1 Marks

The probability of hypothesis H being true given evidence X.

The probability of hypothesis H being true.

The probability of evidence X being observed.

The probability of evidence X being observed given hypothesis H.

5. What does the recall of a classifier measure?

1 Marks

The percentage of actual positive instances correctly identified.

The percentage of predicted positive instances that are actually positive.

The total number of true positives.

The total number of false negatives.

6. What is a decision tree used for in Machine Learning?

1 Marks

Data cleaning

Predicting categorical class labels

Predicting numerical values

Data visualization

7. What does a leaf node in a decision tree represent?

1 Marks

A class label

An attribute

A test on an attribute

A split in the data

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Information gain

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Gain ratio

Entropy

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Decreased accuracy

Improved classification performance

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Binary split

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Singular Split

None of the options given

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1 Marks

Evidence

Prior probability

Likelihood

Posterior probability

14. What is overfitting in a classification model?

1 Marks

When the model performs well on training data but poorly on unseen data

When the model performs well on all data

When the model uses too few attributes

When the model is too simple

15. Formula for bayes theorem: $P(H|X) =$

1 Marks

$(P(X|H) * P(H)) / P(X)$

$(P(X|H) + P(H)) / P(X)$

$(P(X|H) * P(X)) / P(H)$

None of the options given

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$((1+\beta^2) * \text{precision} * \text{recall}) / (\beta^2 * \text{precision} + \text{recall})$

$((1+\beta) * \text{precision} * \text{recall}) / (\beta^2 + \text{precision} + \text{recall})$

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17. What does the Gini index measure in classification?

1 Marks

Class label accuracy

Impurity of a data partition

Information gain

Prior probability

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1 Marks

Boosting gives more weight to misclassified instances

Boosting combines models with equal weights

Boosting reduces model size

Boosting uses only one model

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1 Marks

Interior

Posterior

Prior

Exterior

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1 Marks

Instance-based learning

Supervised learning

Reinforcement learning

Unsupervised learning

22. What is the effect of increasing k in KNN?

1 Marks

Reduces overfitting and smoothens decision boundaries

Makes the model more complex

Increases model variance

Reduces model accuracy

23. What is the primary reason for using slack variables in SVC?

1 Marks

To allow misclassification of some data points in non-linearly separable data

To reduce the number of support vectors

To remove noise from the dataset

To make the model deterministic

24. If two classes in a dataset are highly overlapping, which SVC kernel is likely to perform best?

1 Marks

No kernel will work

Linear kernel

Polynomial

RBF

25. Which factor has the most significant impact on KNN's performance?

1 Marks

The activation function used

The number of support vectors

The number of output classes

The choice of k value and distance metric

26. What is the key difference between AdaBoost and Bagging?

1 Marks

AdaBoost assigns different weights to samples, while Bagging uses bootstrapping

Bagging focuses on reducing bias, while AdaBoost reduces variance

Bagging uses weak learners, while AdaBoost does not
AdaBoost requires deep learning models

27. How does Boosting train weak learners?

1 Marks

[Sequentially, with each new learner correcting the mistakes of the previous one](#)

Independently and in parallel

By using the same data for each learner

By averaging predictions from multiple models

28. Which Scikit-Learn module contains the Support Vector Classifier (SVC)?

1 Marks

`sklearn.neighbors`

[sklearn.svm](#)

`sklearn.tree`

`sklearn.ensemble`

29. Which Scikit-Learn function is used to generate a confusion matrix?

1 Marks

`sklearn.metrics.accuracy_score`

[sklearn.metrics.confusion_matrix](#)

`sklearn.metrics.classification_report`

`sklearn.metrics.roc_auc_score`

30. You are evaluating a model and see the following two approaches:

from sklearn.model_selection import cross_val_predict # Approach A

model.fit(X, y)

pred_A = model.predict(X) # Approach B

pred_B = cross_val_predict(model, X, y, cv=5)

What is the defining difference between the predictions in pred_A and pred_B?

1 Marks

pred_B predicts probabilities, while pred_A predicts continuous integers.

[In pred_B, each data point's prediction was generated by a model that was never trained on that specific data point.](#)

pred_A is used for Regression, while pred_B is strictly for Classification.

There is no difference; they yield the exact same numerical array.

1. In the context of classification, what is a classifier?

1 Marks

[A model derived from data used to predict class labels.](#)

An algorithm that identifies patterns in data.

A measure of the accuracy of a prediction.

A technique for visualizing data.

2. What theorem forms the basis of Bayesian classification?

1 Marks

Bayes' Theorem

Central Limit Theorem

Law of Large Numbers

Pythagorean Theorem

3. In Naive Bayes, what does $P(H|X)$ represent?

1 Marks

The probability of hypothesis H being true given evidence X.

The probability of hypothesis H being true.

The probability of evidence X being observed.

The probability of evidence X being observed given hypothesis H.

4. What does the precision of a classifier refer to?

1 Marks

The percentage of predicted positive instances that are actually positive.

The percentage of actual positive instances correctly identified.

The total number of true positives.

The total number of false negatives.

5. What is a key characteristic of the Random Forest ensemble method?

1 Marks

It is specifically designed for handling missing values.

It uses a single, highly optimized decision tree.

It does not require any training data.

It combines multiple decision trees, each trained on a different random subset of features.

6. What is the Gini index used for in CART?

1 Marks

Attribute selection

Model evaluation

Data visualization

Predicting numerical values

7. What is the F1-score used for?

1 Marks

Measuring accuracy

Measuring precision only

Measuring recall only

Balancing precision and recall

8. What is a random forest in classification?

1 Marks

A combination of decision trees

A single decision tree

A linear classifier

A single Naive Bayes model

9. Formula for bayes theorem: $P(H|X) =$

1 Marks

$(P(X|H) * P(H)) / P(X)$

$(P(X|H) + P(H)) / P(X)$

$(P(X|H) * P(X)) / P(H)$

None of the options given

10. Give the formula for Accuracy.

1 Marks

$(TP + TN) / (P + N)$

$(TP - TN) / (P + N)$

$(TP + TN) / (P - N)$

None of the options given

11. Give the formula for Sensitivity.

1 Marks

TP / P

TF / F

TP / F

TF / P

12. Give the formula for Specificity.

1 Marks

TF / P

TP / P

TP / F

TN/N

13. Give the formula for Recall.

1 Marks

None of the options given

$TP / (TP - FP)$

$TP / (TP + FP)$

$TP / (TP + FN)$

14. What does a confusion matrix display?

1 Marks

Predicted vs. actual classifications

Accuracy of the model

Model size

Data distribution

15. Which of the following statements about SVC is TRUE?

1 Marks

SVC can use the kernel trick to classify non-linearly separable data

The number of support vectors increases with increasing margin

SVC works only for linearly separable data

SVC cannot handle imbalanced data

16. What is the function of the kernel trick in SVC?

1 Marks

Map the input space into a higher-dimensional space to make data linearly separable

Reduce the number of features

Speed up model training

Reduce overfitting

17. What type of learning does KNN belong to?

1 Marks

Instance-based learning

Supervised learning

Reinforcement learning

Unsupervised learning

18. What is the effect of increasing k in KNN?

1 Marks

Reduces overfitting and smoothens decision boundaries

Makes the model more complex

Increases model variance

Reduces model accuracy

19. What is the primary reason for using slack variables in SVC?

1 Marks

To allow misclassification of some data points in non-linearly separable data

To reduce the number of support vectors

To remove noise from the dataset

To make the model deterministic

20. If two classes in a dataset are highly overlapping, which SVC kernel is likely to perform best?

1 Marks

No kernel will work

Linear kernel

Polynomial

RBF

21. What type of classifiers does AdaBoost primarily use?

1 Marks

Support Vector Machines

Deep neural networks

K-Nearest Neighbors

Weak learners, typically decision stumps

22. What is the key difference between AdaBoost and Bagging?

1 Marks

AdaBoost assigns different weights to samples, while Bagging uses bootstrapping

Bagging focuses on reducing bias, while AdaBoost reduces variance

Bagging uses weak learners, while AdaBoost does not

AdaBoost requires deep learning models

23. What is the main difference between Bagging and Boosting?

1 Marks

Bagging reduces variance, while Boosting reduces bias

Bagging and Boosting both reduce variance

Boosting randomly selects training samples, while Bagging assigns weights to samples

Boosting works only with decision trees, while Bagging does not

24. Which of the following is an example of a Bagging algorithm?

1 Marks

Random Forest

AdaBoost

Gradient Boosting

XGBoost

25. Which Scikit-Learn function is used to generate a confusion matrix?

1 Marks

sklearn.metrics.accuracy_score

`sklearn.metrics.confusion_matrix`

`sklearn.metrics.classification_report`

`sklearn.metrics.roc_auc_score`

26. Which Scikit-Learn function is used to compute the ROC curve?

1 Marks

`sklearn.metrics.roc_curve`

`sklearn.metrics.confusion_matrix`

`sklearn.metrics.precision_recall_curve`

`sklearn.metrics.f1_score`

27. Which Scikit-Learn function is used to visualize a confusion matrix?

1 Marks

`sklearn.metrics.ConfusionMatrixDisplay`

`sklearn.metrics.plot_roc_curve`

`sklearn.metrics.show_confusion_matrix`

`matplotlib.confusion_matrix`

28. Review the following basic instantiation of a Random Forest: `from sklearn.ensemble import RandomForestClassifier rf = RandomForestClassifier(n_estimators=10)`

What does the `n_estimators=10` parameter specifically instruct Scikit-Learn to do?

1 Marks

Restrict the maximum depth of the resulting tree to 10 hierarchical levels.

Use only the 10 most statistically important features from the training dataset.

Train an ensemble consisting of exactly 10 separate decision trees.

Split the dataset into 10 folds for internal cross-validation.

29. Consider a dataset with two features: (Age, ranging from 18 to 65) and (Income, ranging from 20,000 to 150,000). A student runs the following code on the raw data:

```
from sklearn.neighbors import KNeighborsClassifier model =
```

```
KNeighborsClassifier(n_neighbors=3)
```

```
model.fit(X_train, y_train)
```

Because the student skipped using a `StandardScaler`, what will happen to the KNN model's logic?

1 Marks

The model will crash because KNN requires input features to be exactly between 0 and 1.

The model will automatically scale the data internally before calculating distances.

Age () will heavily dominate the predictions because its values are smaller and more dense.

Income () will completely dominate the Euclidean distance calculations, rendering Age almost useless.

30. You are evaluating a model and see the following two approaches:

```
from sklearn.model_selection import cross_val_predict # Approach A
```

```
model.fit(X, y)
```

```
pred_A = model.predict(X) # Approach B  
pred_B = cross_val_predict(model, X, y, cv=5)
```

What is the defining difference between the predictions in pred_A and pred_B?

1 Marks

pred_B predicts probabilities, while pred_A predicts continuous integers.

In pred_B, each data point's prediction was generated by a model that was never trained on that specific data point.

pred_A is used for Regression, while pred_B is strictly for Classification.

There is no difference; they yield the exact same numerical array.